

# Secure Authentication with 3D Manipulation in Dynamic Layout for Virtual Reality

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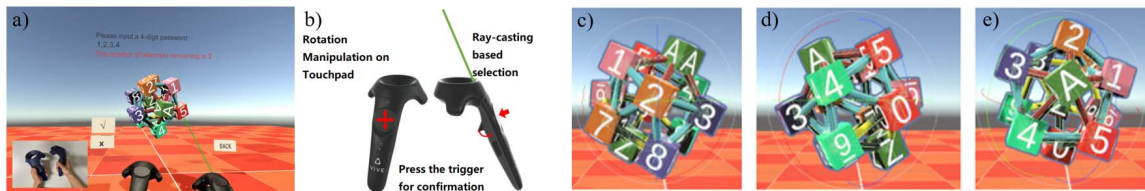


Figure 1: Implementation of the proposed authentication scheme using 3D manipulation in dynamic layout: a) Password entry process, user can manipulate layout for inputting using thumb on the controller touchpad, and select the password number with controller trigger. b) the interaction design based on controllers, including target selection and confirmation, and layout manipulation. c-e) different views of manipulation layout for inputting.

## ABSTRACT

Since Virtual Reality (VR) applications are applied with many traditional fields (e.g., education, game, shopping), the user authentication is necessary to protect privacy and security (e.g., Shoulder Surfing attacks). In this work, we design and implement an interactive sphere-like input layout for user authentication. To enhance the security, the layout of password keeps rotated dynamically, which asks users to enter different characters with slight hand movement changes. Furthermore, the layout can be rotated horizontally via touchpad of the VR controller, which can provide usability while maintaining high security. This research can enrich the knowledge-based authentication in VR.

## 1 INTRODUCTION

Malicious attacks against VR applications may cause serious consequences such as stealing user privacy information. User authentication is one of vital components for VR applications (e.g., online VR shopping, VR social, VR education). Existing studies have shown that traditional digital passwords and two-dimensional graphic authentication schemes cannot effectively resist attackers stealing users' passwords through observation in the real world when deployed in a VR environment [2]. The leakage of passwords may cause serious consequences such as the disclosure of user privacy.

On one hand, the existing knowledge-based authentication methods are error-prone and easy to attack. On the other hand, the biometric authentication methods require external tracking or recognition devices, which brought additional cost and weight for headsets. The limitation of these authentication methods would greatly destroy user experience in VR [3], so it is necessary to design and implement a secure authentication scheme that can be directly deployed in a VR environment.

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This work proposes a new knowledge-based user authentication scheme, we propose an interactive dynamic layout with thumb manipulation on controller touchpads for user authentication. To enhance the security, the layout of password keeps rotated dynamically, which asks users to enter different characters with slight hand movement changes. Furthermore, the layout can be rotated horizontally through controller touchpad, which can provide the usability while maintaining high security.

## 2 RELATED WORK

Commonly used interactive methods of selection include point-and-click using controllers, gaze selection, grasping, touch, and selection using head motion. Recent studies have shown that the use of 3D interactive layouts can improve the level of security, and the corresponding choice of different interaction methods can also affect the security and usability of the authentication schemes. Point-and-click using controllers, gaze selection, and head motion are some of the better interaction methods in terms of usability and observation resistance, with point-and-click using controllers requiring less time and achieving higher accuracy [4]. In this work, we therefore adopt the virtual hand manipulation for authentication to achieve a good balance between security and usability.

Existing research shows that shoulder surfing in real life is the main weakness of knowledge-based authentication schemes [1]. The threat level of shoulder surfing is generally related to the professional level of the attacker who observes in the real world, recording equipment and other conditions. The higher the professional level of the attacker and the more complete the recording method (e.g., using paper, pen, video and other different ways to record the user's input actions), the greater the security threat it poses. In addition, the attacker may record videos of the user's actions and then use techniques such as machine learning to compare them and thus analyze the user's input. Observation resistance is crucial for secure authentication of VR users.

## 3 IMPLEMENTATION

The design of dynamic layout and interaction method during authentication process is shown in Figure 1.

### 3.1 Layout Design

The input layout is designed as a hollow dodecahedron that can be rotated 360 degrees around the X-axis, Y-axis, and Z-axis. There

are 12 vertices in total, and each vertex represents an input character. The key space of this scheme is  $12^n$ , and  $n$  denotes the cipher length. The 3D layout can be rotated autonomously or its specific rotation interaction can be manipulated with controller touchpad by users.

To enhance security, the layout will be randomly deflected by an angle. Since the initial rotation angle of the model is different during each run, in order to facilitate the user to select the object whose initial state is located at the back, the model will automatically rotate along the horizontal axis in a 360-degree cycle at a uniform speed when there is no interactive operation. The time required to automatically rotate one circle is 10 seconds. The sphere-like design enables seamless switching of the main operating surface of the model (i.e., the surface of the model on which the user selects input), while the manual manipulation of its rotation action is designed creatively to be achieved by touching different positions of the touchpad on the controller.

In order to enhance usability and allow the user to see the keys and their corresponding characters in other planes, every two adjacent vertices are connected by a thin rod, which is set to be not selected, and over the thin rod, the user can see the vertices (i.e., input keys) on the back of the model, which facilitates the next interactive operation for input.

### 3.2 Authentication Process

Users first enter the interface, in which they can choose to set the password or perform identity verification for two phases.

First, users need to select the function of password setting to enter the setting interface first, and follow the prompts to set the password, after setting, click the submit button on the setting page to save, if the set password meets the requirements, it can be submitted successfully; otherwise, users need to follow the prompts to modify the password.

Second, after successfully setup the password, the user can choose to enter the authentication interface to complete the authentication function. In the authentication interface, the user needs to enter the saved password first and then click the submit button. If the submitted password is the same as the saved set password, the authentication is passed; otherwise, the user will be prompted to fail the authentication. In addition, if the user can successfully pass within 3 attempts specified by the scheme, the authentication is still counted as successful in the end.

### 3.3 Interaction Consideration

In order to facilitate the user's operation, the input layout is placed in the center of the field of view, so that the user basically does not need to move his head to change the field of view when interacting, and it may not cause dizziness after a long time of use.

To facilitate the user to check whether the input is correct, the password requirement and what the user has entered are displayed at the top of the input model; in addition, the user is prompted with the number of remaining attempts (3 attempts in total), the time spent on input, and whether the authentication is passed or not.

## 4 PERFORMANCE EVALUATION

In order to evaluate the usability and security of this scheme, two additional user authentication schemes were implemented in this experiment (as shown in Figure 2), namely, the traditional two-dimensional PIN password scheme and the three-dimensional tesseract PIN password scheme with depth information added to the former. Then a comparison experiment between this design and these two schemes was conducted. Both schemes have a fixed static keyboard layout and the same key settings of the proposed scheme, and the user could select objects and thus enters passwords by using the same interaction as the scheme presented in this work.

We recruited 20 subjects for this study, more than half of the subjects in the subjective questionnaire thought that this scheme was

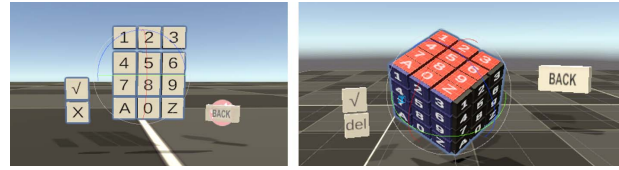


Figure 2: Two existing user authentication schemes as a comparison for evaluation.

fun to interact with, and most of the subjects preferred to use this scheme even though the authentication time was similar with the two fixed layout.

In the security evaluation, the subjects were able to guess the input of the 2D PIN scheme smoothly, but the motivation of the subjects decreased significantly with the difficulty of guessing the password of the 3D PIN scheme and this proposed scheme. The experimental results of the guessing accuracy show that the attacker can achieve the same accuracy rate when guessing the 4-digit password of this scheme as the 6-digit password of the traditional PIN scheme. It indicates that the security of this proposed scheme was better than other twos.

For evaluation of authentication schemes, increased security often means decreased usability, however, this design of touchpad and manipulation achieves a good balance between security and usability.

## 5 CONCLUSION

This work aimed to design and implement a 3D dynamic layout with two hands interaction for authentication in VR. Compared with two traditional schemes, the proposed method allows seamless switching of the main operating surface of the model (i.e., the surface on which the user selects the input), while the manual manipulation of its rotation is creatively designed to be achieved by touching different positions of the touchpad on the controller. The proposed authentication scheme could be a good balance for security and usability and applied to various VR applications (e.g., VR shopping, VR education, VR social, and metaverse).

## ACKNOWLEDGMENTS

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